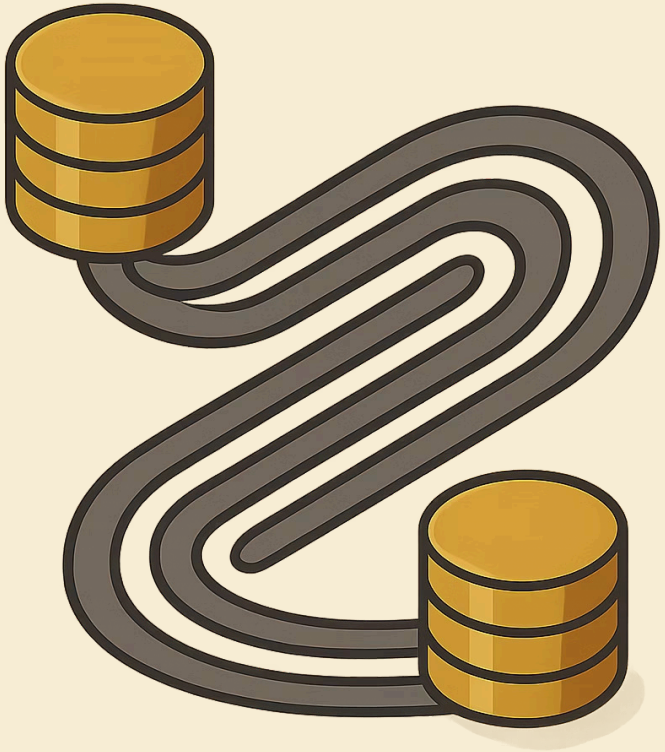

Units I–V: Foundation for Our Capstone Project

How Scala + Spark Powers the Amazon Reviews Data Lakehouse Pipeline

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Unit I – Scala Programming Basics

Core Fundamentals

Variables, data types, conditionals, functions, loops, and REPL environment

Functional Programming

Immutability, higher-order functions, closures, anonymous functions, map, filter, reduce

Object-Oriented

Classes, objects, traits, inheritance, and sbt build environment

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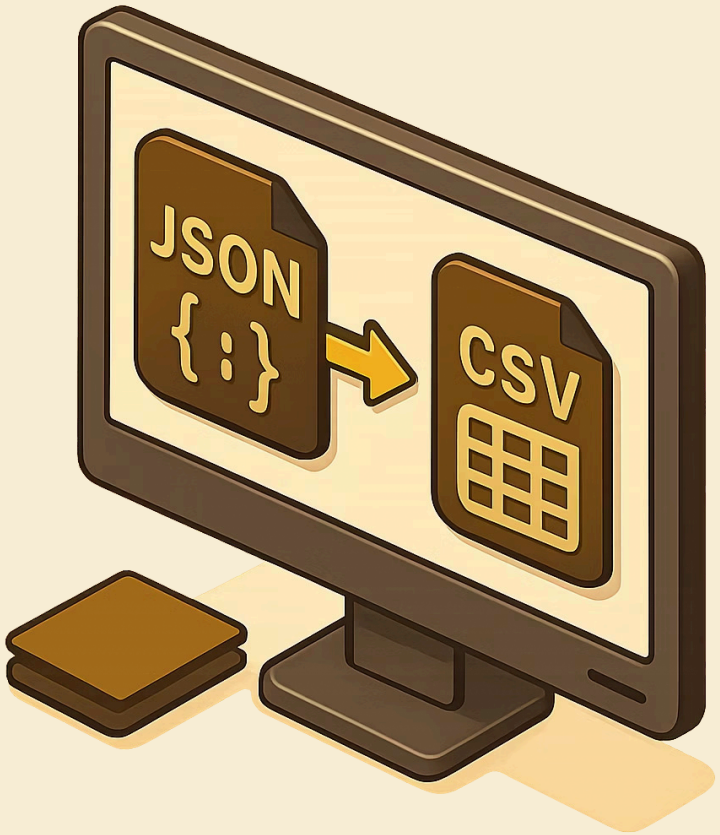

Unit II – Scala for Big Data

Pattern Matching & Safety

Case classes for schemas, Option for safe null handling, and error handling techniques

Collection Mastery

Understanding immutable vs mutable collections and functional transformations

File I/O Operations

Reading and writing JSON and CSV files for data processing

Production Code

Preparing Scala code for Spark applications and real-world big data work

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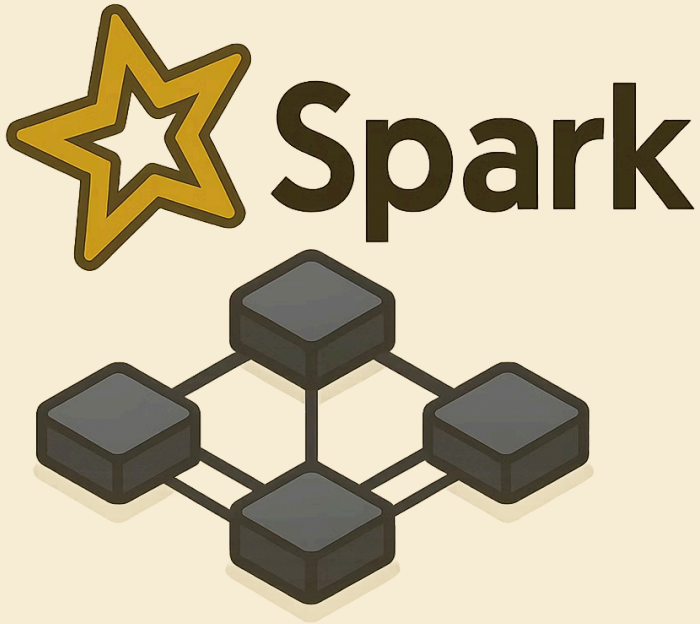

Unit III – Introduction to Apache Spark

Driver Program

Central coordinator that creates SparkContext and submits jobs

Cluster Manager

Resource allocation across nodes (Standalone, YARN, Mesos)

Executors

Worker processes that run tasks and store data in memory

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Unit IV – DataFrames, Datasets & Spark SQL

1

DataFrame API

Schema definition, aggregations, joins, and window functions for structured data processing

2

Dataset API

Compile-time type safety combining benefits of RDDs and DataFrames

3

Spark SQL

SQL queries on DataFrames with Catalyst optimizer for query planning and optimization

4

MLlib Introduction

Pipelines, transformers, estimators, and common machine learning algorithms

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Unit V – Spark Streaming & Performance

Structured Streaming

Continuous processing of data streams with sources, sinks, watermarking, windowing, and checkpointing

Performance Tuning

Partitioning strategies, caching mechanisms, broadcast variables, and accumulator optimization

Production Ready

Industry use cases, optimization techniques, and best practices for real-world deployment

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How All Units Come Together

01

Unit I & II

Scala functional code, case classes, and sbt build environment

02

Unit III

Spark architecture and Delta Lake table management

03

Unit IV

DataFrames, Spark SQL, and MLlib Pipeline construction

04

Unit V

Structured Streaming and performance tuning techniques

Result: End-to-end Medallion Lakehouse combining batch processing, real-time streaming, and machine learning

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Project Architecture

Medallion Architecture with Streaming and ML

Raw CSV Files

Amazon Reviews dataset as source data

Bronze Layer (Delta)

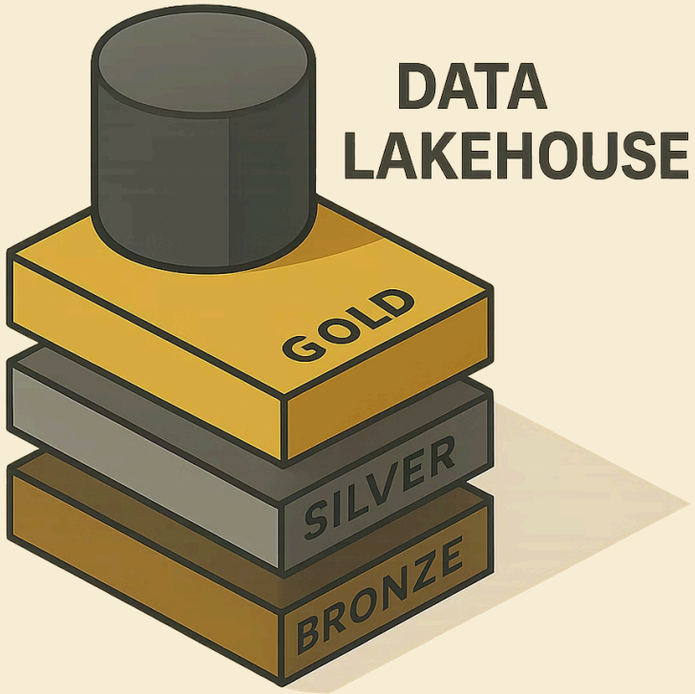
Raw data ingested into Delta Lake format

Silver Layer (Cleaned)

Data quality checks and standardization

Gold Layer (Aggregations)

Business-level aggregations and MLlib Sentiment Model



This is our final architecture. Bronze-Silver-Gold using Delta Lake, plus Structured Streaming for real-time reviews, plus MLlib for sentiment prediction in Gold layer. This is exactly what companies expect from a Data Engineer.

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Learning Outcomes & Job Relevance

Medallion Architecture

Full implementation with Delta Lake for reliable data management

Production Stack

Scala, Spark, MLlib, and Structured Streaming in action

End-to-End Pipeline

Complete ETL, real-time processing, and machine learning integration

Portfolio Ready

Ready for Databricks and Data Engineer roles with proven skills

By the end of this capstone you will have a complete, portfolio-ready project that covers every CO2 to CO6. You can write on your resume: 'Built end-to-end Spark Scala Lakehouse with Medallion Architecture, Structured Streaming and MLlib for Amazon Reviews dataset'.

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Today's Agenda – Let's Build

1. **Project Setup**
   build.sbt configuration and SparkSession with Delta
2. **Bronze Layer**
   CSV to Delta Lake ingestion
3. **Silver + Gold**
   Data cleaning and ML Pipeline construction
4. **Streaming Intro**
   Real-time processing foundation

Theory is done. Now we start heavy practical coding. We will build the first working layers today using your downloaded Amazon Reviews data. Everyone code along on Lubuntu VirtualBox.

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